

Ruprecht-Karls-Universität Heidelberg
Institut für Informatik

Bachelor Thesis
Parallel ASP Planning

Name: Levi Maximilian Klein
Student ID: 4283957
Supervisor: Dr. Gnad
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Ich versichere, dass ich diese Bachelor-Arbeit selbstständig verfasst und nur die angegebenen Quellen und Hilfsmittel verwendet habe.

Heidelberg, dd.mm.yyyy

Abstract

This is the abstract.

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1 Introduction

Automated planning is a fundamental area of Artificial Intelligence that deals with finding a sequence of actions to transition a system from an initial state to a desired goal state. Finding an efficient and effective plan is crucial in various applications and industries such as robotics and logistics. In general, planning problems are computationally challenging, often falling into NP-hard or even PSPACE-complete complexity classes. Therefore, developing efficient algorithms and representations for planning tasks to reduce the search space and computation time is of great importance.

1.1 Motivation

1.2 Goal of the Thesis

1.3 Structure of the Thesis

Here you describe the structure of the thesis. For example: In chapter 2, we formally define the notion of planning tasks, first describing sequential plans and afterwards three different kinds of parallel plans. We then present the ASP encodings for the different plans and proofing their soundness. In chapter ??e show the pipeline from translating a pddl problem to validating the given plan and introduce different optimization strategies

2 Theory

2.1 Definition

Definition 1 *Multi-valued planning task*. A STRIPS-like multivalued planning task according to (Helmert 2006) is given by a 4 tuple $\langle \mathcal{F}, s_0, s_*, \mathcal{O} \rangle$, where:

- $\mathcal{F}$ is a finite set of state variables, also called **fluents**. Each $x \in \mathcal{F}$ has a finite domain and in every state, x takes exactly one value from its domain.
- s_0 , the **initial state**, is a total function, s.t. $s_0(x) \in \text{dom}(x)$ for each $x \in \mathcal{F}$, i.e. a function that maps each fluent to a value from its domain.
- s_* , the **goal conditions**, is a partial function, s.t. $s_*(x) \in \text{dom}(x)$, for each $x \in \text{dom}(s_*)$ i.e. a function that specifies the desired values for a subset of fluents.
- $\mathcal{O}$ is a finite set of operators, also called **actions**. We define an action a as $\langle a^c, a^e \rangle$, where a^c is the precondition and a^e is the postcondition of a . Both a^c and a^e are partial functions, s.t. $a^c(x) \in \text{dom}(x)$ for each $x \in \text{dom}(a^c)$ and $a^e(x) \in \text{dom}(x)$ for each $x \in \text{dom}(a^e)$.

Remark. An action a also has a finite set of effects eff , where $eff = \langle cond, x, v \rangle$. $cond$, the effect condition, is a (possibly empty) partial function, s.t. $cond(x) \in \text{dom}(x)$ for each $x \in \text{dom}(cond)$. $x \in \mathcal{F}$ is the effected fluent and $v \in \text{dom}(x)$ is the value assigned to x if $cond$ holds.

Example 1. Let us illustrate the definitions with a small example. The plan is to leave the house by opening the door, going outside, and finally closing the door.

- $\mathcal{F} = \{door, location\}$ with $\text{dom}(door) = \{open, closed\}$, $\text{dom}(location) = \{inside, outside\}$
- Initial state: $s_0 = \{door=closed, location=inside\}$
- Goal: $s_* = \{door=closed, location=outside\}$
- Actions: $\mathcal{O} = \{openDoor, closeDoor, goOut\}$, where

$$openDoor = \langle \{door=closed\}, \{door=open\} \rangle$$

$$closeDoor = \langle \{door=open\}, \{door=closed\} \rangle$$

$$goOut = \langle \{location=inside, door=open\}, \{location=outside\} \rangle$$

Definition 2 *Successor state*. Let s be state and $a \in \mathcal{O}$ an action.

The *successor state* $o(a, s)$, obtained by applying action a in state s , is defined if the precondition

of a holds, i.e., $a^c(x) = s(x)$ for all $x \in \text{dom}(a^c)$. Then

$$s'(x) = o(a, s) = \begin{cases} a^e(x) & \text{if } x \in \text{dom}(a^e) \\ s(x) & \text{otherwise.} \end{cases}$$

Remark. Let $A_n = \langle a_1, \dots, a_n \rangle$ be a sequence of actions with length n . We define the application of A_n to a state s recursively by

$$o(A_n, s) = o(a_n, o(A_{n-1}, s))$$

if $o(A_i, s)$ is defined for all $1 \leq i \leq n$.

Definition 3 Sequential plan. A *sequential plan* is a sequence of actions $A_n = \langle a_1, \dots, a_n \rangle$, s.t. $s' = o(A_n, s_0)$ is defined and $s'(x) = s_*(x)$ for all $x \in \text{dom}(s_*)$.

Example 2. Consider the planning task from the previous example. One possible sequential plan is $A = \langle \text{openDoor}, \text{goOut}, \text{closeDoor} \rangle$. Since

$$\begin{aligned} o(A, s_0) &= o(\langle \text{openDoor}, \text{goOut}, \text{closeDoor} \rangle, \{\text{door}=\text{closed}, \text{location}=\text{inside}\}) \\ &= o(\langle \text{goOut}, \text{closeDoor} \rangle, \{\text{door}=\text{open}, \text{location}=\text{inside}\}) \\ &= o(\text{closeDoor}, \{\text{door}=\text{open}, \text{location}=\text{outside}\}) \\ &= \{\text{door}=\text{closed}, \text{location}=\text{outside}\} = s_* \end{aligned}$$

For parallel actions and plans, we first need the definition of a confluent set of actions.

Definition 4 confluent. Let $A = \{a_1, \dots, a_n\} \subseteq \mathcal{O}$ be a set of actions.

A is *confluent* : $\iff a_i^e(x) = a_j^e(x)$ for all $1 \leq i < j \leq n$ and each $x \in \text{dom}(a_i^e) \cap \text{dom}(a_j^e)$

Remark. A is confluent $\iff B$ is confluent for all $B \subseteq A$

Example 3. In our example $\{\text{openDoor}, \text{goOut}\}$ and $\{\text{closeDoor}, \text{goOut}\}$ are confluent sets. We extend the example with the fluents *light* and *key* where $\text{dom}(\text{light}) = \text{dom}(\text{key}) = \{0, 1\}$ and two actions *lightsOut* and *getKey*:

$$\begin{aligned} \text{lightsOut} &= \langle \{\text{location}=\text{inside}, \text{light}=1\}, \{\text{light}=0\} \rangle \\ \text{getKey} &= \langle \{\text{location}=\text{inside}, \text{key}=0\}, \{\text{key}=1\} \rangle \end{aligned}$$

Additionally, we modify the precondition: $\text{openDoor}^c = \{\text{door}=\text{closed}, \text{key}=1\}$ and our new goal is $s_* = \{\text{location}=\text{outside}, \text{door}=\text{closed}, \text{light}=0\}$.

Now for example $\{\text{getKey}, \text{openDoor}, \text{goOut}, \text{lightsOut}\}$ and $\{\text{getKey}, \text{closeDoor}, \text{goOut}, \text{lightsOut}\}$ are also confluent sets of actions.

Definition 5 *Parallel representations*. Let s be a state and $A = \{a_1, \dots, a_n\}$ be a confluent set of actions. Then A is

- (i) $\forall$ -step serializable in $s : \iff o(\langle a_{\pi(1)}, \dots, a_{\pi(n)} \rangle, s)$ is defined for all permutations π
- (ii) $\exists$ -step serializable in $s : \iff o(\langle a_{\pi(1)}, \dots, a_{\pi(n)} \rangle, s)$ is defined for some permutation π
and $a^c(x) = s(x)$ for each $a \in A$ and $x \in \text{dom}(a^c)$
- (iii) relaxed $\exists$ -step serializable in $s : \iff o(\langle a_{\pi(1)}, \dots, a_{\pi(n)} \rangle, s)$ is defined for some
permutation π

Remark. A is $\forall$ -step serializable $\Rightarrow A$ is $\exists$ -step serializable $\Rightarrow A$ is relaxed $\exists$ -step serializable.

Definition 6 *Parallel Plans*. Let $A = \langle A_1, \dots, A_n \rangle$ be a sequence of confluent sets of actions. A is a $\forall$ -step, $\exists$ -step or relaxed $\exists$ -step plan iff

- $s_n(x) = s_*(x)$ for each $x \in \text{dom}(s_*)$
- A_i is $\forall$ -step, $\exists$ -step or relaxed $\exists$ -step serializable in s_{i-1} for $1 \leq i \leq n$, where

$$s_i(x) = \begin{cases} a^e(x) & \text{if } x \in \text{dom}(a^e) \\ s_{i-1}(x) & \text{else} \end{cases}$$

for each $a \in A_i$.

Example 4. In our modified example we have:

- $\forall$ -step plan: $\langle \{getKey, lightsOut\}, \{openDoor\}, \{goOut\}, \{closeDoor\} \rangle$
- $\exists$ -step plan: $\langle \{getKey, lightsOut\}, \{openDoor\}, \{goOut, closeDoor\} \rangle$
- relaxed $\exists$ -step plan: $\langle \{getKey, lightsOut, openDoor, goOut\}, \{closeDoor\} \rangle$

2.2 Introduction to ASP

Answer Set Programming (ASP) is a declarative problem-solving approach oriented towards difficult (primarily NP-hard) search problems. Unlike imperative programming, where the programmer specifies the control flow to compute a solution, ASP follows a declarative paradigm: the problem is described as a logic program such that its valid models—called answer sets—correspond to the solutions of the original problem. This methodology allows for a clean separation between the problem description (the logic program) and the solving algorithms (implemented by ASP solvers like clingo).

The theoretical foundation of ASP lies in the stable model semantics of logic programming (Gelfond and Lifschitz 1988). To formally define these programs, we first introduce their basic building blocks. The most fundamental units are atoms, which represent elementary propositions that can be either true or false. These atoms are combined to form literals. A literal is defined as either an atom A (a positive literal) or its default negation *not* A (a negative literal). Here,

the connective *not* denotes negation as failure, meaning that not A is considered true if no proof for A can be derived from the program.

Logic Programs

Based on these definitions, we can define the structure of the programs used to model planning tasks. A normal logic program consists of a finite set of rules, facts, and integrity constraints. Rules of a normal program have the following form:

$$\textbf{Fact: } A_0. \quad (2.1)$$

$$\textbf{Rule: } A_0 :- L_1, \dots, L_n. \quad (2.2)$$

$$\textbf{Integrity Constraint: } :- L_1, \dots, L_n. \quad (2.3)$$

The head of a rule or fact is an atom, which has the same syntactic form as a constant or a function, i.e. $prec(X, V, t)$. The body consists of a (possible empty) list of literals $L_1, \dots, L_n$. Each L_i is of the form A or *not* A , where A is an atom and *not* the default negation. The $:-$ operator can be read as $\Leftarrow$, therefore a rule can be interpreted as "if all literals in the body are true, then the head is true". A fact is a rule without a body, meaning that the head atom is unconditionally true. An integrity constraint is a rule without a head, that filters solution candidates, meaning that all the literal in its body must not be satisfied, i.e. at least one literal in the body must be false. A set of atoms is called a *model* of a logic program, if all facts, rules and integrity constraints are satisfied. Atoms are considered true if they are in the model, otherwise they are considered false. In ASP, a model is called an *answerset* or *stablemodel*, if every atom in the model has an acyclic derivation of the program.

```

1  fluent(door; loc; light; key).
2  value(door, 0).  value(door, 1).
3  value(loc, in).  value(loc, out).
4  value(light, 0). value(light, 1).
5  value(key, 0).  value(key, 1).
6
7  init(door, 0). init(loc, in).  init(light, 1). init(key, 0).
8  goal(door, 0). goal(loc, out). goal(light, 0).
9
10 action(openDoor; closeDoor; goOut; lightOut; getKey).
11 prec(openDoor, door, 0).  prec(openDoor, key, 1).  post(openDoor, door, 1).
12 prec(closeDoor, door, 1).  post(closeDoor, door, 0).
13 prec(goOut, loc, in).  prec(goOut, door, 1).  post(goOut, loc, out).
14 prec(lightOut, loc, in).  prec(lightOut, light, 1).  post(lightOut, light, 0).
15 prec(getKey, loc, in).  prec(getKey, key, 0).  post(getKey, key, 1).

```

Listing 1: ASP encoding of the example planning task

Combining these facts with the right encodings, stable models correspond to the *sequential*, $\forall$ -step, $\exists$ -step or *relaxed* $\exists$ -step plans.

```

1  #include <incmode>.
2  holds(X, V, 0) :- init(X, V).
3  #program check(t).
4  :- query(t), goal(X, V), not holds(X, V, t).
5  #program step(t).
6  {holds(X, V, t) : value(X, V)} = 1 :- fluent(X).
7  {occurs(A, t)} :- action(A).
8  :- occurs(A, t), post(A, X, V), not holds(X, V, t).
9  :- holds(X, V, t), not holds(X, V, t-1), not occurs(A, t) : post(A, X, V).

```

Listing 2: Core for sequential and parallel encodings for planning

All planning algorithms share a common incremental encoding using the incmode for clingo as seen in Listing 2 (Gebser et al. 2014). The program is divided into three subprograms: *base*, *step(t)* and *check(t)*.

The *base* subprogram contains the static part of the encoding, which is only grounded once. Line 2 defines the initial state s_0 , where the predicate $holds(X, V, t)$ is used to represent fluents $X \in \mathcal{F}$ that have value $V \in \text{dom}(X)$ at timestep t .

$check(t)$ is a parametrized subprogram starting at $t = 0$. The integrity constraint in Line 4 checks if $s_*(X) = s_t(X)$ for each $X \in \text{dom}(s_*)$. $Query(t)$ is true only for the latest integer t . $step(t)$ is grounded after $check(t)$ fails for each timestep t starting at $t = 1$. Note that postconditions of applied actions in timestep t are also set in t and the preconditions have to hold in $t - 1$. Lines 6 defines holds for times step t , giving fluent X one value V from its domain and $occurs(A, t)$ in Line 7 are all possible actions in timestep t obeying rules 8 and 9. Line 8 ensures that the postcondition of an action is applied and also assert the confluence of any set of action applied in parallel since if $V = a^e(X) \neq \tilde{a}(X) = \tilde{V}$ either $holds(X, V, t)$ or $holds(X, \tilde{V}, t)$ would be false. Line 9 makes sure that fluents not affected by any action keep their value from the previous timestep.

So far, the common core handles the transitions from one state to the next in regards of the postcondition of applied actions.

2.2.1 encoding for sequential plans

```

1 :- occurs(A, t), prec(A, X, V), not holds(X, V, t-1).
2 :- #count{A : occurs(A, t)} > 1.
```

Listing 3: Extension for sequential plans

Starting with the extension for sequential plans as seen in Listing 3, Line 2 ensures that the precondition of an applied action is met in $t - 1$ and Line 3 only allows one action to occur per timestep.

Example 5. Running this with our example, we get three possible sequential plans.

```

Answer: 1
occurs(getKey,1) occurs(lightOut,2) occurs(openDoor,3) occurs(goOut,4) occurs(closeDoor,5)
Answer: 2
occurs(getKey,1) occurs(openDoor,2) occurs(lightOut,3) occurs(goOut,4) occurs(closeDoor,5)
Answer: 3
occurs(lightOut,1) occurs(getKey,2) occurs(openDoor,3) occurs(goOut,4) occurs(closeDoor,5)
```

Listing 4: Output from clingo for sequential encoding

2.2.2 encoding for $\forall$ -step plans

```

1 :- occurs(A, t), prec(A, X, V), not holds(X, V, t-1).
2 :- occurs(A, t), prec(A, X, V), not post(A, X, _), not holds(X, V, t).
3 single(X, t) :- occurs(A, t), prec(A, X, V1), post(A, X, V2), V1 != V2.
4 :- single(X, t), #count{A : occurs(A, t), post(A, X, V)} > 1.

```

Listing 5: Extension for $\forall$ -step plans

Now for the extension for $\forall$ -step plans as seen in Listing 5, Line 1 is the same as in the sequential encoding. As mentioned before, the integrity constraint in Line 8 of the common core already ensures confluence of applied actions. Since the sequence of the actions has to be valid for all permutations, we have to make sure that no action invalidates the precondition of any other action. Let a be an action applied in parallel and $X \in \text{dom}(a^c)$. There are two cases to consider:

1. the fluent X remains unchanged, i.e. $X \in \text{dom}(a^c) \cap \text{dom}(a^e)$ and $a^c(X) = a^e(X)$ or $X \in \text{dom}(a^c)$ and $X \notin \text{dom}(a^e)$
2. the postcondition of a invalidates its precondition, i.e. $X \in \text{dom}(a^e) \cap a^c(X) \neq a^e(X)$ for an action a

Line 2 handles the first case, while the second case is a little trickier. If there is such an action, there can not be another action $\tilde{a}$ with $X \in \text{dom}(\tilde{a}^e)$, since if $a^e(X) = \tilde{a}^e(X)$, the precondition of a is invalidated and if $a^e(X) \neq \tilde{a}^e(X)$, the confluence is undermined. For this, we define these actions described in (ii) as $single(X, t)$ in Line 3 and the integrity constraint in Line 4 ensures that there is at most one such action for each fluent X in timestep t .

Example 6. Running this with our example, we get two possible $\forall$ -step plans.

```

Answer: 1
occurs(getKey,1) occurs(openDoor,2) occurs(lightOut,2) occurs(goOut,3) occurs(closeDoor,4)
Answer: 2
occurs(lightOut,1) occurs(getKey,1) occurs(openDoor,2) occurs(goOut,3) occurs(closeDoor,4)

```

Listing 6: Output from clingo for $\forall$ -step encoding

2.2.3 encoding for $\exists$ -step plans

```

1 :- occurs(A, t) , prec(A, X, V), not holds(X, V ,t -1).
2 apply(A1, t) :- action(A1),
3     ready(A2, t) : post(A1, X, V1), prec(A2, X, V2), A1 != A2, V1 != V2.
4 ready(A, t) :- action(A), not occurs(A, t).
5 ready(A, t) :- apply(A, t).
6 :- action(A) , not ready(A, t).

```

Listing 7: Extension for $\exists$ -step plans

```

1 :- occurs(A, t), prec(A ,X ,V), not holds(X, V, t-1).
2 #edge((A1, t), (A2, t)): occurs(A1, t),
3     post(A1, X, V1), prec(A2, X, V2), A1 != A2, V1 != V2.

```

Listing 8: Alternative Extension for $\exists$ -step plans

There are two different ways to encode the extension for $\exists$ -step plans as seen in Listing 7 and Listing 8. As before, Line 1 in Listing 7 and 8 is the same as in the sequential and $\forall$ -step encoding. To ensure that there is at least one valid permutation, either an action a_1 that invalidates the precondition of another action a_2 has to be applied after a_2 or action a_2 does not occur in this timestep. For this, all actions has to be marked as *ready* (Line 6). An action is *ready* if it either does not occur in this timestep (Line 4) or if it is captured by *apply* (Line 2). An action can be safely applied, if all other actions whose preconditions it invalidates are ready, i.e. are either applied beforehand or not occurring in this timestep. Finally, Line 6 ensures that actions does not circularly invalidates each other.

Example 7. Running this with our example, we get three possible $\exists$ -step plans.

```

Answer: 1
occurs(getKey,1) occurs(openDoor,2) occurs(closeDoor,3) occurs(goOut,3) occurs(lightOut,3)
Answer: 2
occurs(getKey,1) occurs(openDoor,2) occurs(lightOut,2) occurs(closeDoor,3) occurs(goOut,3)
Answer: 3
occurs(lightOut,1) occurs(getKey,1) occurs(openDoor,2) occurs(closeDoor,3) occurs(goOut,3)

```

Listing 9: Output from clingo for $\exists$ -step encoding

2.2.4 encoding for relaxed $\exists$ -step plans

```

1 reach(X, V, t) :- holds(X, V, t - 1).
2 reach(X, V, t) :- occurs(A, t), apply(A, t), post(A, X, V).
3 apply(A1, t) :- action(A1), reach(X, V, t) : prec(A1, X, V);
4     ready(A2, t) : post(A1, X, V1), prec(A2, X, V2), A1 != A2 , V1 != V2.
5 ready(A, t) :- action(A), not occurs(A, t).
6 ready(A, t) :- apply(A, t).
7 :- action(A), not ready(A, t).

```

Listing 10: Extension for relaxed $\exists$ -step plans

Finally, the extension for relaxed $\exists$ -step plans as seen in Listing 10 is quite similar to the $\exists$ -step encoding. The difference between those two is that the precondition of all actions do not have to hold simultaneously but they can "help" each other out. This is done by defining the predicate `reach/3` in Line 1 and 2 as all fluents that can be reached by the step before or applied action in this timestep. An action can then be safely applied, if its precondition is reachable and as before, all actions that it invalidates are ready (Line 4). Again Line 7 ensures that applied action do not circularly invalidate each other.

Example 8. Running this with our example, we get three possible relaxed $\exists$ -step plans.


```

Answer: 1
occurs(openDoor,1) occurs(getKey,1) occurs(closeDoor,2) occurs(goOut,2) occurs(lightOut,2)
Answer: 2
occurs(openDoor,1) occurs(lightOut,1) occurs(getKey,1) occurs(closeDoor,2) occurs(goOut,2)
Answer: 3
occurs(openDoor,1) occurs(goOut,1) occurs(lightOut,1) occurs(getKey,1) occurs(closeDoor,2)

```

Listing 11: Output from clingo for relaxed $\exists$ -step encoding

2.3 Proof of Correctness

In order to prove the correctness of the presented encodings, we define the following:

- B is the rule in Line 1 of Listing 3.
- $Q(t)$ is the integrity constraint in the check program directive (Line 5) of Listing 2.
- $S(t)$ are the rules and integrity constraint in the step program directive (Lines 6-9) of Listing 2.
- $S^p(t)$ is the extension for the specific plans, where $p = s$ for Listing 3, $p = \forall$ for Listing 5, $p = \exists$ for Listing 7, $p = E$ for Listing 8 and $p = R$ for Listing 10.
- I is the set of facts representing a planning task $\langle \mathcal{F}, s_0, s_*, \mathcal{O} \rangle$.
- $\langle a_1, \dots, a_n \rangle$ is a sequence of actions.
- $\langle A_1, \dots, A_m \rangle$ is a sequence of sets of actions.
- $P^p(n) := I \cup B \cup Q(0) \cup \bigcup_{t=1}^n (Q(t) \cup S(t) \cup S^p(t)) \cup \{query(n)\}$, where $p \in \{s, \forall, \exists, E, R\}$.

Theorem 1 (Correctness of sequential plans (DIMOPOULOS et al. 2019)).

$\langle a_1, \dots, a_n \rangle$ is a sequential plan $\iff P^s(n)$ has a stable model M , s.t.

$\{\langle a, t \rangle \mid occurs(a, t) \in M\} = \{\langle a_t, t \rangle \mid 1 \leq t \leq n\}$.

Proof. ($\Rightarrow$): Let $\langle a_1, \dots, a_n \rangle$ be a sequential plan.

First we show that for any stable model M of $P^s(n)$ s.t. $\{\langle a, t \rangle \mid occurs(a, t) \in M\} = \{\langle a_t, t \rangle \mid 1 \leq t \leq n\}$, the rules and integrity constraints ensures that $\{\langle X, V, t \rangle \mid holds(X, V, t) \in M\} = \{\langle X, V, t \rangle \mid X \in \mathcal{F}, V = o(\langle a_1, \dots, a_t \rangle, s_0), 0 \leq t \leq n\}$.

- (i) the rule in B ensures that $holds(X, V, 0) = s_0(X)$ for each $X \in \mathcal{F}$
- (ii) for any $1 \leq t \leq n$, both integrity constraints in $S(t)$ ensure that $holds(X, V, t) = a_t^e(X)$ if $X \in \text{dom}(a_t^e)$ and $holds(X, V, t) = s_{t-1}(X)$ otherwise.

Therefore, the only stable model M that $P^s(n)$ can have is

$$\begin{aligned}
 M = & I \cup \{occurs(a_t, t) \mid 1 \leq t \leq n\} \\
 & \cup \{holds(X, V, t) \mid X \in \mathcal{F}, v = o(\langle a_1, \dots, a_t \rangle, s_0), 0 \leq t \leq n\} \cup \{query(n)\}.
 \end{aligned}$$

s.t. $\{\langle a, t \rangle \mid \text{occurs}(a, t) \in M\} = \{\langle a_t, t \rangle \mid 1 \leq t \leq n\}$.

One can easily check that all rules and integrity constraints in $P^s(n)$ are satisfied and M is indeed a stable model of $P^s(n)$.

($\Leftarrow$): Let M be a stable model of $P^s(n)$ s.t. $\{\langle a, t \rangle \mid \text{occurs}(a, t) \in M\} = \{\langle a_t, t \rangle \mid 1 \leq t \leq n\}$.

We proof by induction over t , that for all $1 \leq t \leq n$:

$$\{\langle X, V \rangle \mid \text{holds}(X, V, t-1) \in M\} = \{\langle X, V \rangle \mid X \in \mathcal{F}, V = o(\langle a_1, \dots, a_{t-1} \rangle, s_0)(X)\}.$$

I-Start: $t=0$

The Rule in B implies: $\{\langle X, V \rangle \mid \text{holds}(X, V, 0) \in M\} = \{\langle X, V \rangle \mid X \in \mathcal{F}, s_0(X) = V\}$.

I-Step: $t \mapsto t+1$

The integrity constraint in Line 1 of Listing 3 in $S^s(t)$ ensures that the successor state of $s(t)$, i.e. $o(a_{t+1}, o(\langle a_1, \dots, a_t \rangle, s_0))$ is defined. Additionally the choice rule in Line 6 of Listing 2 and the integrity constraints in $S(t)$ guarantee that

$$\{\langle X, V \rangle \mid \text{holds}(X, V, t+1) \in M\} = \{\langle X, V \rangle \mid X \in \mathcal{F}, V = o(\langle a_1, \dots, a_{t+1} \rangle, s_0)(X)\}.$$

Finally, with the fact $\text{query}(n)$, the integrity constraint in $Q(n)$ yields that $o(\langle a_1, \dots, a_n \rangle, s_0)(X) = s_*(X)$ for all $X \in \text{dom}(s_*)$. Thus, $\langle a_1, \dots, a_n \rangle$ is a sequential plan. $\square$

Theorem 2 (Correctness of parallel plans (DIMOPOULOS et al. 2019)).

$\langle A_1, \dots, A_m \rangle$ is a $\forall$ -exists-/relaxed exists-step plan $\iff P^p(m)$ has a stable model M s.t., $\{\langle a, t \rangle \mid \text{occurs}(a, t) \in M\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$ with $p \in \{\forall, \exists, R\}$.

Proof. ($\Rightarrow$) $\langle A_1, \dots, A_m \rangle$ is a $\forall$ -exists-/relaxed exists-step plan.

For each $1 \leq t \leq m$, let s_t be the state s.t. $s_t(X) = \begin{cases} a^e(X) & \forall a \in A_t \text{ and } X \in \text{dom}(a^e) \\ s_{t-1}(X) & \forall X \in \mathcal{F} \setminus \bigcup_{a \in A_t} \text{dom}(a^e) \end{cases}$.

Then, as in the proof above, for any stable model M of $P^p(m)$ s.t. $\{\langle a, t \rangle \mid \text{occurs}(a, t) \in M\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$, the rule in B together with the choice rule and the integrity constraints in $S(t)$ make sure that $\{\langle X, V, t \rangle \mid \text{holds}(X, V, t) \in M\} = \{\langle X, V, t \rangle \mid X \in \mathcal{F}, V = s_t(X), 0 \leq t \leq m\}$. We now split the proof into the different cases.

Case 1: $\langle A_1, \dots, A_m \rangle$ is a $\forall$ -step plan.

The atom $\text{single}(X, t)$ is derived by the rule in $S^\forall(t)$ iff there exists some action $a \in A_i$ s.t. $X \in \text{dom}(a^c) \cap \text{dom}(a^e)$ and $a^c(X) \neq a^e(X)$ for $1 \leq t \leq m$. In other words, if there is an action that invalidates its own precondition. Hence $P^\forall(m)$ cannot have any stable model M apart

from

$$\begin{aligned}
M = & I \cup \{occurs(a, t) \mid a \in A_t, 1 \leq t \leq m\} \\
& \cup \{holds(X, V, t) \mid X \in \mathcal{F}, V = s_t(X), 0 \leq t \leq m\} \\
& \cup \{single(X, t) \mid X \in \text{dom}(a^c) \cap \text{dom}(a^e), a^c(X) \neq a^e(X), a \in A_t, 1 \leq t \leq m\} \\
& \cup \{query(m).\}
\end{aligned}$$

s.t. $\{\langle a, t \rangle \mid occurs(a, t) \in M\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$. Since for any $1 \leq t \leq m$, the successor state $s_t = o(\langle a_1, \dots, a_k \rangle, s_{t-1})$ is defined for any sequence $\langle a_1, \dots, a_k \rangle$ s.t. $\{a_1, \dots, a_k\} = A_t$, we have that M satisfies all integrity constraints in $S^\forall(t)$. As above, one can easily check that all other rules and integrity constraints in $P^\forall(m)$ are satisfied as well. Therefore M is a stable model of $P^\forall(m)$.

Case 2: $\langle A_1, \dots, A_m \rangle$ is a $\exists$ -step plan.

Then for any $1 \leq t \leq m$, there is a sequence of actions $\langle a_1, \dots, a_k \rangle$, s.t. $\{a_1, \dots, a_k\} = A_t$ and $a_i^e(X) = \tilde{a}_j^e(X)$ if $X \in \text{dom}(a_i^e) \cap \text{dom}(\tilde{a}_j^e)$ for any $1 \leq i < j \leq m$. That is the edges of the directed graph:

$$G_t = (\mathcal{O}, \{\langle a, \tilde{a} \rangle \mid a \in A_t, \tilde{a} \in \mathcal{O} \setminus \{a\}, X \in \text{dom}(a^e) \cap \text{dom}(\tilde{a}^e), a^e(X) \neq \tilde{a}^e(X)\})$$

can connect some a_i for $1 \leq i \leq k$ to elements of $\mathcal{O} \setminus \{a_i, \dots, a_k\}$ only if G_t remains acyclic. The acyclicity of G_t yields that the recursive derivation of $apply(a, t)$ and $ready(a, t)$ is successful, i.e. they both belong to the least fixpoint in $S^\exists(t)$ (Line 2-5 in 7) for each $a \in \mathcal{O}$ and $1 \leq t \leq m$ and thus, $P^\exists(m)$ cannot have any stable model apart from

$$M' = M \cup \{apply(a, t) \mid a \in \mathcal{O}, 1 \leq t \leq m\} \cup \{ready(a, t) \mid a \in \mathcal{O}, 1 \leq t \leq m\}$$

s.t. $\{\langle a, t \rangle \mid occurs(a, t) \in M'\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$. As above, one can easily check that all other rules and integrity constraints in $P^\exists(m)$ are satisfied as well. Therefore M is a stable model of $P^\exists(m)$.

Case 3: $\langle A_1, \dots, A_m \rangle$ is a *relaxed* $\exists$ -step plan. Then for any $1 \leq t \leq m$. there is a sequence of actions $\langle a_1, \dots, a_k \rangle$ s.t. $\{a_1, \dots, a_k\} = A_t$ and $o(a_i, \langle a_1, \dots, a_{i-1} \rangle, s_{t-1})$ is defined for $1 \leq i \leq k$, which implies that $a_i^e(X) = a_j^e(X)$ if $X \in \text{dom}(a_i^e) \cap \text{dom}(a_j^e)$ for $i < j \leq k$. Given this, the least fixpoint of $S^R(t)$ (Lines 1-6 in Listing 10) contains:

- (i) $reach(X, V, t)$ for each $X \in \mathcal{F}$ and $V \in \{s_{t-1}(X), s_t(X)\}$
- (ii) $apply(a, t)$ for each $a \in \mathcal{O}$ s.t. $a^c(X) \in \{s_{t-1}, s_t\}$ and all $X \in \text{dom}(a^c)$
- (iii) $ready(a, t)$ for each $a \in \mathcal{O}$

Thus, the only stable model $PR(m)$ can have is

$$\begin{aligned}
 M = & I \cup \{occurs(a, t) \mid a \in A_t, 1 \leq t \leq m\} \\
 & \cup \{holds(X, V, t) \mid X \in \mathcal{F}, s_t(X) = V, 0 \leq t \leq m\} \\
 & \cup \{reach(X, V, t) \mid X \in \mathcal{F}, V \in \{s_{t-1}, s_t\}, 1 \leq t \leq m\} \\
 & \cup \{apply(a, t) \mid a \in \mathcal{O} : a^c(X) \in \{s_{t-1}(X), s_t(X)\} \forall X \in \text{dom}(a^c)\} \\
 & \cup \{ready(a, t) \mid a \in \mathcal{O}, 1 \leq t \leq m\} \\
 & \cup \{query(m)\}
 \end{aligned}$$

s.t. $\{\langle a, t \rangle \mid occurs(a, t) \in M'\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$. As above, one can easily check that all other rules and integrity constraints in $PR(m)$ are satisfied as well. Therefore M is a stable model of $PR(m)$.

($\Leftarrow$) M is a stable model of $PP(m)$ s.t. $\{\langle a, t \rangle \mid occurs(a, t) \in M'\} = \{\langle a, t \rangle \mid a \in A_t, 1 \leq t \leq m\}$. For any $1 \leq t \leq m$, the choice rule in Line 6 of Listing 2 makes sure that s_t s.t. $s_t(X) = V$ iff $holds(X, V, t) \in M$ is a state. Moreover, the rule in B together with the integrity constraints in $S(t)$ yield that $s_t(x) = a^e(X)$ for each $a \in A_t$ and $X \in \text{dom}(a^e)$, which also implies that A_t is confluent, and $s_t(X) = s_{t-1}(X)$ for each $X \in \mathcal{F} \setminus \bigcup_{a \in A_t} \text{dom}(a^e)$. With the fact $query(n)$, the integrity constraint in $Q(n)$ yields that $s_m(X) = s_*(X)$ for all $X \in \text{dom}(s_*)$. We now split the proof into the different cases, to show that A_t is $\forall$ -/ $\exists$ -/*relaxed* $\exists$ -step serializable in s_{t-1} .

Case 1: To show: A_t is $\forall$ -step serializable.

Let $\langle a_1, \dots, a_k \rangle$ be a sequence of actions s.t. $\{a_1, \dots, a_k\} = A_t$. We proof by induction, that for all $1 \leq i \leq k$: $o(\langle a_1, \dots, a_i \rangle, s_{t-1})$ is defined.

I-Start: $i = 1$

The integrity constraint in Line 1 of Listing 5 guarentees that $o(a_i, s_{t-1})$ is defined for each $a_i \in A_t$.

I-Step: $i \mapsto i + 1$

For any $X \in \text{dom}(a_{i+1}^c) \cap \bigcup_{1 \leq j \leq i+1} \text{dom}(a_j^e)$, we have that $o(\langle a_1, \dots, a_i \rangle, s_{t-1})(X) = s_t(X)$.

We now look at two cases:

- (i) $X \notin \text{dom}(a_{i+1}^e)$: The integrity constraint in Line 2 of Listing 5 yields that $s_t(X) = a_{i+1}^c(X) = s_{t-1}(X)$.
- (ii) $X \in \text{dom}(a_{i+1}^e)$: The integrity constraint in Line 5 of Listing 5 implies that $single(X, t) \notin M$, which means that the rule of $single(X, t)$ is not satisfied and $s_t(X) = a_{i+1}^e(X) = a_{i+1}^c(X) = s_{t-1}(X)$ must be the case.

This shows that $o(\langle a_1, \dots, a_i \rangle, s_{t-1})(X) = a_{i+1}^c(X)$, so that $o(\langle a_1, \dots, a_{i+1} \rangle, s_{t-1})$ is defined. Therefore $o(\langle a_1, \dots, a_k \rangle, s_{t-1})$ is defined.

Case 2: To show: A_t is $\exists$ -step serializable.

As in Case1, the integrity constraint in Line 1 of Listing 7 and 8 ensures that for any $a \in A_t$, $o(a, s_{t-1})$ is defined. The integrity constraint in Line 6 of Listing 7 implies that $ready(a, t) \in M$

for each $a \in \mathcal{O}$. The rules in Line 5 and 6 further yield that $apply(a, t) \in M$ for each $a \in A_t$. Given the rule in Lines 2-3, for any $1 \leq i \leq k$, there must be some $a_i \in A_t$, s.t. $\{a_j \in A_t \mid i < j \leq k, X \in \text{dom}(a_i^c) \cap \text{dom}(a_j^e), a_i^e(X) \neq a_j^e(X)\} = \emptyset$. In turn, we have that $a_j^e(X) = a_i^e(X) = s_{t-1}(X)$ if $X \in \text{dom}(a_i^c) \cap \text{dom}(a_j^e)$ for $1 \leq j \leq i$, so that $o(\langle a_1, \dots, a_i \rangle, s_{t-1})$ and, in particular, $o(\langle a_1, \dots, a_k \rangle, s_{t-1})$ is defined.

Case 3: To show: A_t is *relaxed* $\exists$ -step serializable.

In view of the rule in Line 1 of Listing 10, we have that $reach(X, V, t) \in M$ if $s_{t-1}(X) = V$. The integrity constraint in Line 7 further implies that $ready(a, t) \in M$ for each $a \in \mathcal{O}$ and the rules in Line 5 and 6 yield that $apply(a, t) \in M$ for each $a \in A_t$. Given the rules in Lines 2 and 3-4, for any $1 \leq i \leq k$, there must be some $a_i \in A_t$, s.t. $\{a_j \in A_t \mid i < j \leq k, X \in \text{dom}(a_j^c) \cap \text{dom}(a_i^e), a_i^e(X) \neq a_j^e(X)\} = \emptyset$ and $a_i^e(X) \in \{s_{t-1}\} \cup \{a_j^e(X) \mid 1 \leq j \leq i, X \in \text{dom}(a_j^e)\}$ for all $X \in \text{dom}(a_i^c)$. Hence, for each $X \in \text{dom}(a_i^c)$, we have that $a_i^e(X) = s_{t-1}(X)$ and $X \notin \bigcup_{1 \leq j \leq i} \text{dom}(a_j^e)$ or $a_j^e(X) = a_i^e(X)$ if $X \in \text{dom}(a_j^e)$, for $1 \leq j < i$, so that $o(\langle a_1, \dots, a_i \rangle, s_{t-1})$ and, in particular, $o(\langle a_1, \dots, a_k \rangle, s_{t-1})$ is defined. $\square$

3 Method

3.1 Pipeline Overview

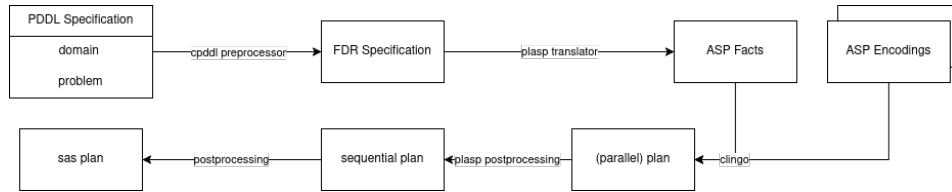


Figure 3.1: Pipeline for testing the different planning encodings

To test and compare the different parallel planning encodings against each other and against the sequential planning encoding, we set up a python pipeline using the lab framework (Seipp et al. 2017) and the unofficial collection of (sequential) IPC benchmark instances from the downward project ¹.

Starting at a planning task in the Planning Domain Definition Language (PDDL), we first preprocess it using cpddl (Fišer 2025) that converts it into a finite domain representation (FDR). Plasp then takes the FDR as input and translates it into ASP facts representing the planning task. Further information of this step can be found in (DIMOPOULOS et al. 2019). Using clingo with a modified version of the incremental encodings presented above together with the facts from the output of plasp, we can then compute sequential, $\forall$ -step, $\exists$ -step or relaxed $\exists$ -step plans as stable models. Finally, to validate the given plan, we postprocess the parallel representation back into a sequential plan, using the postprocess encoding provided by plasp and validate the plan with VAL ². Each step has some optimizations and configurations that will be explained in the following sections.

3.2 Changes in the ASP encodings

To use the presented encodings in our pipeline, we had to make some small changes to the original encodings from (DIMOPOULOS et al. 2019).

¹<https://github.com/aibasael/downward-benchmarks>

²<https://github.com/KCL-Planning/VAL>

```

1  #include <incmode>.
2  #program base.
3  fluent(X) :- variable(X).
4  fluent(X,V) :- initialState(X,V), fluent(X).
5  value(X,V) :- contains(X, V).
6  prec(A, X, V) :- precondition(A, X, V).
7  post(A, X, V) :- postcondition(A, _, X, V).
8  ...
9  :- fluent(X), #count{V : holds(X,V,0)} > 1.
10 ...
11 #program step(t).
12 ...
13 #show.
14 #show occurs(A, t): occurs(A, t).

```

Line 3-6 just renames the predicates from plasp's output to match the ones used in the encodings, while Line 7 uses the wildcard `_` to ignore the effect condition. This let clingo assume that all effects are fulfilled, which reduces the search space but might lead to not fully optimal plans. Line 9 ensures that each fluent has at most one value in the initial state and finally Line 13-14 specify which atoms should be shown in the output.

3.3 Optimizations

3.3.1 Mutexes

Definition 7 *mutex and mutex groups*. A set of facts $M \subseteq \mathcal{F}$ is called

- (i) a *mutex* $\iff$ for all reachable states $s : M \not\subseteq s$.
- (ii) a *mutex group* $\iff$ for all reachable states $s : |M \cap s| \leq 1$.

In other words, a mutex is a set of fluents that can never be true at the same time in any reachable state, while a mutex group is a set of fluents where at most one fluent can be true at the same time. Mutexes and mutex groups can be used to optimize the planning encodings by reducing the search space. For this, we extend our encoding in Listing 2 with the following rules:

```

1      #program base.
2      ...
3      mutex(G,X) :- mutexGroup(G), contains(G,X,V), fluent(X,V).
4      mutex(G)   :- mutexGroup(G), #count{X : mutex(G,X)} > 1.
5      :- mutex(G), #count{X,V : holds(X,V,0), contains(G,X,V)} > 1.^
6      ...
7      #program step(t).
8      ...
9      :- mutex(G), #count{X,V : holds(X,V,t), contains(G,X,V)} > 1.
10     ...

```

Listing 12: Expansion for mutexes

Here, the mutex group G is given by plasp as facts of the form *mutexGroup*/1 and *contains*/3. In the base program, we first define which fluents belong to a mutex group (Line 3), where a group must contain at least two fluents (Line 4). The integrity constraint in Line 5 ensures that no more than one fluent of each mutex group holds in the initial state and the integrity constraint in Line 9 ensures the same for each state in timestep t .

3.3.2 Guess and Check

An optimization specific to the (relaxed) $\exists$ -step encodings is the guess and check approach. The general idea is to first guess a solution without looking for actions that circularly invalidate each other and afterwards check if the guessed solution has a cycle. Formally, we encode the problem by a pair of programs $\langle G, C \rangle$ where G is the guessing program and C is the checking program. Stable models M of G yields a solution if $C \cap M$ is *unsatisfiable*. For making "systematic" guesses, (DIMOPOULOS et al. 2019) proposes to use the core program ($S(t)$) in Listing 2 together with the integrity constraint in Line 1 of Listing 3, 5 or 7 (B) and the facts representing a planning task. Thus, a stable model M of G is s.t. the pre- and postconditions hold for all actions applied in parallel (Line 8 of Listing 2 and Line 1 of Listing 3, 5 or 7). and the integrity constraint in Line 8 of Listing 2 also ensures confluence of applied actions. The only thing remained to be checked is if actions circularly invalidate each other, i.e. $a^e(X) \neq \tilde{a}^e(X)$ and $\tilde{a}^e(\tilde{X}) \neq \tilde{a}^e(\tilde{X})$. This can be accomplished by taking the atoms over *occurs*/2 from M as facts together with a program C containing the facts representing a planning task and the following encoding:


```

1  #include <incmode>.
2  #program check(t).
3  :- query(t), not cycle(t).
4  #program step(t).
5  prec(A, X, V, t) :- prec(A, X, V), occurs(A, t).
6  post(A, X, V, t) :- post(A, X, V), occurs(A, t).
7  action(A, t) :- occurs(A, t).
8  apply(A1, t) :- action(A1, t),
9      ready(A2, t) :- post(A1, X, V1, t), prec(A2, X, V2, t), A1 != A2 , V1 != V2.
10 ready(A, t) :- action(A, t), not occurs (A, t).
11 ready (A, t) :- apply(A, t).
12 cycle(t) :- action (A, t), not ready (A, t).

```

Listing 13: Encoding for checking circular invalidation

Here, the maximum value for t in subprogram *check* and *step* is $m = \max\{t \mid \text{occurs}(a, t)\}$ and C becomes *unsatisfiable* if $t > m$. Like in Listing 7, we check for each t if all actions $A \in \text{occurs}(A, t)$ can be applied by the predicate *ready* (Line 7-11). If actions in A circularly invalidate each other, *ready*(A, t) is underivable for these actions and thus *cycle*(t) (Line 12) in step t is derived. Hence, C becomes satisfiable if there is a circular invalidation in step t , which means that there is no permutation of the actions A , s.t. the successor state is defined and therefore not (relaxed) $\exists$ -step serializable.

3.3.3 Heuristic

The general idea of heuristics in clingo described by (DIMOPOULOS et al. 2019) is to extend the search heuristic by giving each atom a

- *level* ($\in \mathbb{Z}$, 0 by default), allowing to modify the order of which atoms are selected first
- *sign* ($\in \mathbb{Z}$, 0 by default, where values > 0 corresponds to *true*, < 0 to *false* and 0 to *undefined*), which allows for controlling the truth value assigned to atoms.

During solving, these values get modified during the activation of **#heuristic** directives. The search heuristic of clingo selects the atom with the highest level and sets it to its associated *sign*, if *sign* defined, else switching to the default *sign* heuristic.

```

1  #program step(t).
2  #heuristic holds(X, V, t-1):    holds(X, V, t). [2147483647-t, true]
3  #heuristic holds(X, V, t-1):    not holds(X, V, t). [2147483647-t, false]

```

Listing 14: backward propagating heuristic

The heuristic in Listing 14 aims starting at the goal of a planning task and then searching backwards. Here both lines have the level $l = 2^{31} - 1 - t$ where $2^{31} - 1$ is the largest integer supported by clingo, giving the starting steps the highest priority. The second line expresses that whenever clingo made $holds(X, V, t)$ true, $holds(X, V, t - 1)$ should be made true at the level l as well. Similarly, the directive in Line 3 is activated when an instance of $holds(X, V, t)$ becomes false, in which case $holds(X, V, t - 1)$ should be made false at level l as well. Both directives associates atoms over holds/3 with higher level the earlier the state is, i.e. the value for t is smaller, which intends to bias the search of *clingo* to establish the goal conditions as early as possible.

3.4 Experiment

To compare the different planning encodings and the effectiveness of the optimizations, we run four experiments with the pipeline described above. Each experiment run on the same set of benchmark instances with the sequential, $\forall$ -step, *relaxed* $\exists$, *guess and check* and both versions of the $\exists$ -step encodings. The Guess and Check algorithm and the fallback described above are started one after the other with the time limit of 30 minutes, where the times are added together if the check fails and the $\forall$ -step encoding finds a solution. If the summed up time is below the time limit, the problem is considered solved.

Aborted runs are considered unsolved and assigned the time nan. *Cpddl* is equipped with and without the h2 option that extends the search for mutexes and the planning encodings with and without the backward propagating heuristic. The experiments were run on the bwiclust3.0 server of the Karlsruhe Institute of Technology (KIT) using nodes with Intel Xeon Gold 6248R CPUs (3.0 GHz, 24 cores). Each run was limited to one CPU core, where *clingo* and *cpddl* were given a time limit of 30 minutes and a memory limit of 32 GiB each.

4 Results

4.1 One to One Comparison

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